

A Scalable Monte-Carlo / CVaR Linear-Program Optimiser for Mental-Account Portfolios with Many Securities and Derivatives

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Addendum to the 2012 MSc thesis (USI Lugano) and the accompanying paper
on the *Beyond Mean-Variance* optimiser

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Abstract

The thesis and its software reimplementations solve the Das–Statman mental-account problem on an exhaustive return grid. That representation is exact for small universes but its state space grows as m^N in the number of securities N , which confines it in practice to four or five assets and a single derivative. This addendum sets out a *scenario-based* reformulation whose cost is linear in N and independent of how returns are discretised, and which admits an arbitrary number K of derivatives priced consistently inside each scenario. Joint returns are sampled through a copula; each mental account’s constraint—maximise expected return subject to a tail-risk floor—is then a *linear program* via the Rockafellar–Uryasev representation of Conditional Value-at-Risk. We give the full formulation, its relationship to the grid method and to the thesis, a convergence and validation check, and the design trade-offs. A prototype reproduces the closed-form Normal Expected Shortfall to within 3 basis points and optimises a five-asset, two-derivative portfolio in a single linear solve.

1 Motivation: the curse of dimensionality

The Das, Markowitz, Scheid and Statman (2010) mental-account framework—extended in the 2012 thesis with derivatives and structured products—maximises a sub-portfolio’s expected return subject to a behavioural shortfall constraint, $\mathbb{P}(r < H) \leq \alpha$. The engine evaluates this by discretising each security’s return into m levels and enumerating the *joint* state space. For N securities that space has on the order of m^N cells: with $m = 51$, $N = 3$ gives $\sim 1.3 \times 10^5$ states (tractable), whereas $N = 10$ gives $51^{10} \approx 10^{17}$ (infeasible). This is why exhaustive grid search is capped at $N \leq 4$ in the software and why a single derivative is the practical limit.

Institutional portfolios contain tens to hundreds of positions and multiple hedging or yield-enhancement overlays. We therefore need a representation whose cost grows *linearly*, not exponentially, in N , and which scales in the number of derivatives.

2 A scenario-based reformulation

Rather than enumerate a grid, draw a fixed number S of joint *scenarios* from the return distribution. Each scenario s is one simultaneous realisation of all assets,

$$R_s = (R_{s,1}, \dots, R_{s,N}, R_{s,N+1}, \dots, R_{s,N+K}) \in \mathbb{R}^{N+K},$$

where the first N entries are security returns and the last K are derivative returns; every scenario carries equal probability $1/S$. The data structure is an $S \times (N + K)$ matrix, so memory and

per-iteration cost are $O(S(N + K))$: *linear* in the number of instruments, with S chosen for statistical accuracy and entirely decoupled from N . Adding a security widens each scenario by one column; adding a derivative widens it by one more. Nothing is enumerated.

3 Generating the scenarios

Securities. Let $\mu \in \mathbb{R}^N$ and $\sigma \in \mathbb{R}^N$ be the horizon expected returns and volatilities and Ω the correlation matrix, with Cholesky factor L ($\Omega = LL^\top$). Drawing $\xi_s \stackrel{\text{iid}}{\sim} \mathcal{N}(0, I_N)$ and setting $z_s = L\xi_s$, $R_{s,i} = \mu_i + \sigma_i z_{s,i}$ reproduces the thesis’s multivariate-Normal assumption exactly. To capture *tail dependence* (assets falling together in stress), replace the Gaussian copula by a Student- t copula: draw a multivariate- t vector $t_s = z_s/\sqrt{g_s}$ with $g_s \sim \chi_\nu^2/\nu$, map to uniforms $u_{s,i} = F_{t_\nu}(t_{s,i})$, and back to the chosen marginals $z_{s,i} = \Phi^{-1}(u_{s,i})$. The dependence is then heavy-tailed while the marginals remain controlled. This is the Sklar (1959) decomposition into marginals and copula, standard in dependence modelling (Embrechts *et al.*, 2002).

Derivatives (arbitrage-consistent, per scenario). For each derivative k written on underlying $u(k)$, its entry price is the Black–Scholes fair value π_k at $S_0 = 1$ (the same convention as the engine: strikes are fractions of the underlying, volatility is that of the underlying). In scenario s the terminal spot is $1 + R_{s,u(k)}$, the payoff $\Pi_k(1 + R_{s,u(k)})$ follows from the instrument’s leg structure, and the derivative’s scenario return is

$$R_{s,N+k} = \frac{\Pi_k(1 + R_{s,u(k)}) - \pi_k(1 + p_k)}{\pi_k(1 + p_k)},$$

with p_k an optional structured-product premium. Because the payoff is computed from the *same* simulated underlying that drives the security column, each scenario keeps a coherent (underlying, derivative) pair with no auxiliary model. Crucially this holds for *several* derivatives at once, possibly on different underlyings: each simply contributes one more column.

4 The optimisation problem

For weights $w \in \mathbb{R}^{N+K}$ the portfolio return in scenario s is the inner product $r_s = w^\top R_s$, giving S weight-dependent realisations. The expected return is the sample mean $\hat{\mu}^\top w$ with $\hat{\mu} = \frac{1}{S} \sum_s R_s$.

We measure tail risk by Expected Shortfall (Conditional Value-at-Risk), the coherent risk measure (Artzner *et al.*, 1999) and the natural convex counterpart of the thesis’s loss-probability constraint. Writing the loss as $\ell = -r$ and letting α be the tail probability, the Rockafellar–Uryasev (2000, 2002) identity is

$$\text{CVaR}_\alpha(\ell) = \min_{\zeta \in \mathbb{R}} \left\{ \zeta + \frac{1}{\alpha} \mathbb{E}[(\ell - \zeta)^+] \right\},$$

where the minimiser ζ^* is the Value-at-Risk. On the scenario sample this becomes $\zeta + \frac{1}{\alpha S} \sum_s (\ell_s - \zeta)^+$, which is *piecewise-linear and convex*. The tail-average return is $\text{ES}_\alpha(r) = -\text{CVaR}_\alpha(\ell)$, so a behavioural floor “the average outcome in the worst α tail is at least L ” is $\text{CVaR}_\alpha(\ell) \leq -L$.

The linear program

Introduce ζ (scalar) and slacks $z_s \geq 0$. The mental-account problem

$$\max_w \mathbb{E}[r] \quad \text{s.t.} \quad \text{ES}_\alpha(r) \geq L, \quad \mathbf{1}^\top w = 1, \quad w \geq 0$$

becomes the linear program

$$\begin{array}{ll}
\max_{w, \zeta, z} & \hat{\mu}^\top w \\
\text{s.t.} & \zeta + \frac{1}{\alpha S} \sum_{s=1}^S z_s \leq -L, \\
& z_s \geq -w^\top R_s - \zeta, \quad s = 1, \dots, S, \\
& z_s \geq 0, \quad \mathbf{1}^\top w = 1, \quad 0 \leq w \leq \bar{w}.
\end{array}$$

The decision vector $x = (w, \zeta, z) \in \mathbb{R}^{N+K+1+S}$ has $S+1$ inequality rows; the constraint matrix is sparse (each scenario row touches the $N+K$ weights, ζ and one z_s). Modern simplex/interior-point solvers (e.g. HiGHS) handle $N+K \sim 10^2$ and $S \sim 10^4$ in well under a second. Convexity removes the need for the differential-evolution and pattern-search heuristics the grid engine falls back on beyond four assets. Optional bounds $\bar{w}$ (position limits), group budgets, turnover or cardinality penalties are additional linear rows.

5 Value-at-Risk via a CVaR proxy

The thesis constraint $\mathbb{P}(r < H) \leq \alpha$ is a *chance constraint*: non-convex and combinatorial (mixed-integer in general). Rather than solve it exactly, we use CVaR as a conservative convex surrogate. Because $\text{CVaR}_\alpha \geq \text{VaR}_\alpha$ always, imposing a CVaR floor guarantees the VaR-style floor is met at least as safely; ES is also the coherent measure and the one favoured by modern regulation. Where an exact $\mathbb{P}(r < H) \leq \alpha$ solution is required, the convex CVaR optimum is an excellent warm start for a subsequent mixed-integer refinement.

6 Multiple derivatives and mental accounts

Two structural generalisations fall out directly:

- **Several derivatives.** K derivative columns are optimised jointly at no extra modelling cost; the optimiser can, for instance, fund downside protection on the riskiest holding while gearing upside on another. The current engine, by contrast, prices one derivative on one underlying.
- **Mental accounts.** Each behavioural sub-portfolio (goal) is the same linear program over the *same* scenario matrix with its own triple (H_j, α_j, L_j) . Solving the accounts independently and aggregating preserves the goals-based interpretation while inheriting the scalability.

7 Efficient frontier and warm-starting

Sweeping the floor L traces the return/tail-risk frontier. Consecutive programs differ only in the single right-hand side $-L$, so each solve is warm-started from the previous optimal basis and resolves almost instantly; the whole frontier costs little more than one solve. Using common random numbers (the *same* scenario draw across the sweep) removes Monte-Carlo noise from the *shape* of the frontier.

8 Convergence, variance reduction and validation

Scenario estimates carry sampling error of order $S^{-1/2}$; tail quantities such as ES need more scenarios than the distribution's body, so antithetic variates, importance sampling of the loss tail, and common random numbers are part of the engine rather than afterthoughts.

Prototype check. A reference implementation¹ on a five-asset Gaussian universe ($\alpha = 5\%$, $S = 4 \times 10^4$) reproduces the closed-form Normal Expected Shortfall of an equal-weight portfolio,

$$\text{ES}_\alpha = \mu_p - \sigma_p \frac{\phi(\Phi^{-1}(\alpha))}{\alpha},$$

to 3 basis points (Table 1), returns a monotone frontier whose realised ES sits exactly on each floor, correctly reports infeasibility when no portfolio can reach a floor, and solves a two-derivative ($K=2$) instance in a single linear program. A Student- t copula deepens the equal-weight tail from -18.70% to -19.68% , as intended.

Quantity	Monte-Carlo	Closed form	$ \Delta $
Equal-weight $\mathbb{E}[r]$	0.0810	0.0820	0.0010
Equal-weight σ	0.1301	0.1305	0.0004
Equal-weight $\text{ES}_{5\%}$	-0.1870	-0.1873	0.0003

Table 1: Monte-Carlo estimates against closed-form Normal values ($S = 4 \times 10^4$). Frontier solutions (omitted) place the realised ES on the imposed floor to machine tolerance.

The exactness of the grid method on small universes gives a second, independent test: the scenario optimum should converge to the grid optimum on the three- and four-asset thesis cases, which is the recommended acceptance test before deploying the engine at $N = 100$.

9 Relationship to the existing method and limitations

The scenario engine *complements* rather than replaces the grid. The grid is exact and deterministic for small N and remains the reference there; the scenario engine is approximate but scales, and is the only viable route for large, multi-derivative portfolios. The principal cautions are: (i) Monte-Carlo error, mitigated by sample size and variance reduction; (ii) *scenario quality*—the copula choice and the estimation of μ, σ, Ω drive the result, and estimation error worsens with N , so shrinkage or robust estimators become part of the pipeline; and (iii) the CVaR-for-VaR surrogacy described in §5. None of these is a barrier to the construction; each is a calibration responsibility.

10 Commercial relevance

The reformulation turns an academically faithful but small- N tool into one usable on realistic mandates: dozens to hundreds of instruments, several derivatives priced consistently, a coherent tail-risk constraint, and a fast, warm-started frontier. The differentiators are scenario quality, per-scenario arbitrage-consistent derivative pricing, robustness to estimation error, and the goals-based mental-account framing inherited from the thesis.

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¹`mc_cvar_optimizer.py`, accompanying this note: scenario generator, per-scenario derivative pricing, the CVaR linear program and a frontier sweep, in roughly 250 lines of Python/SciPy.

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